

Solving the Game

Finding MI6's Optimal Strategy (p_t, p_f)

1 What Kind of Equilibrium Is This?

It is tempting to call this a *Nash equilibrium* problem, since MI6 and Cheka are clearly playing against one another. But a Nash equilibrium requires **both** players to be choosing a strategy, with each strategy being a best response to the other. That is not the situation here.

Cheka has no strategic freedom in this version of the game. Cheka is told p_t exactly, and it is committed, by construction, to two mechanical steps:

1. Form the exact Bayesian posterior over the three cities, using the known p_t .
2. Search the city with the **maximum** posterior probability.

There is no room in this rule for Cheka to “decide” anything — given a message, its search action is a fixed, deterministic function. Cheka is not optimizing; it is simply computing.

MI6, on the other hand, has two genuine free parameters: p_t (how often to tell the truth) and p_f (how often to follow its own report). MI6 can choose these to shape Cheka's belief however it likes, *because MI6 knows in advance exactly how Cheka will react to any message it sends*.

So what are we actually solving?

This is a **one-sided constrained optimization** — closer to a *Stackelberg* setup (a leader who moves first and knows the follower's exact reaction function) than to a Nash equilibrium. We are finding:

$$(p_t^*, p_f^*) = \arg \max_{p_t, p_f \in [0,1]} \mathbb{E}[\text{MI6's score} \mid p_t, p_f, r]$$

subject to Cheka's response being the fixed rule above. It ensures MI6 cannot do any better *given that Cheka will always behave exactly this way* — nothing more, nothing less. We are not claiming Cheka is playing optimally for itself; we are simply taking its rule as given (as the problem specifies) and asking what the best counter-strategy is.

2 Cheka's Decision Rule

Cheka starts with a uniform prior ($\frac{1}{3}$ each). Upon receiving message m , its posterior is built from the likelihood

$$\text{likelihood}(m) = p_t, \quad \text{likelihood}(\text{other two}) = \frac{1 - p_t}{2} \text{ each.}$$

Because the prior is uniform, it cancels out of Bayes' rule entirely, and the posterior equals the likelihood directly:

$$\text{posterior}(m) = p_t, \quad \text{posterior}(\text{each other city}) = \frac{1 - p_t}{2}.$$

2.1 The Trust Threshold

Cheka searches the message location if and only if its posterior there exceeds the posterior of the alternatives:

$$\begin{aligned}
 p_t &> \frac{1 - p_t}{2} \\
 2p_t &> 1 - p_t && \text{(multiply both sides by 2)} \\
 3p_t &> 1 && \text{(add } p_t \text{ to both sides)} \\
 p_t &> \frac{1}{3}
 \end{aligned}$$

This gives Cheka exactly two regimes:

- **Trust regime** ($p_t > \frac{1}{3}$): Cheka always searches *the message location*.
- **Distrust regime** ($p_t < \frac{1}{3}$): Cheka always searches *one of the other two*, uniformly.

3 Setting Up MI6's Expected Score

Since “both find it” pays MI6 only 3 (instead of 10), we can write MI6's expected score compactly as

$$\mathbb{E}[\text{MI6}] = 10 \underbrace{P(\text{MI6 finds it})}_{\mu} - 7 \underbrace{P(\text{both find it})}_{P(\text{both})},$$

which follows from $10 P(\text{MI6 alone}) + 3 P(\text{both}) = 10 [P(\text{MI6 alone}) + P(\text{both})] - 7 P(\text{both}) = 10\mu - 7P(\text{both})$.

MI6's own accuracy is simple, directly from the definition of p_f :

$$\mu = P(\text{MI6 finds it}) = p_f \cdot r + (1 - p_f) \cdot \frac{1 - r}{2}.$$

The harder part is $P(\text{both})$ — it requires tracking how MI6's search (driven by p_f , off the report R) and Cheka's search (driven by p_t , off the message M , which is itself derived from R) correlate through their shared root cause, the report R .

4 Deriving $P(\text{both})$

4.1 Trust Regime ($p_t > \frac{1}{3}$): Cheka always searches = M

We need $P(S_{\text{mi6}} = L, M = L)$. Condition on the report R , since S_{mi6} and M are independently randomized once R is fixed:

Case $R = L$ (probability r)

$P(S_{\text{mi6}} = L \mid R = L) = p_f$ (MI6 follows its report).

$P(M = L \mid R = L) = p_t$ (MI6 is truthful).

Contributes: $r \cdot p_f \cdot p_t$.

Case $R \neq L$ (probability $1 - r$, split over 2 wrong reports)

Given $R = X \neq L$: $S_{\text{mi6}} = L$ only through the “don’t follow” branch landing on L — probability $\frac{1-p_f}{2}$. Similarly, $M = L$ only through bluffing landing on L — probability $\frac{1-p_t}{2}$.
 Contributes: $(1 - r) \cdot \frac{1 - p_f}{2} \cdot \frac{1 - p_t}{2}$.

$$P(\text{both} \mid \text{trust}) = r p_f p_t + \frac{(1 - r)(1 - p_f)(1 - p_t)}{4}$$

4.2 Distrust Regime ($p_t < \frac{1}{3}$): Cheka always avoids M

Here S_{cheka} is uniform over the two cities $\neq M$. So $S_{\text{cheka}} = L$ requires *both* $M \neq L$ (else L isn’t even a candidate) *and* the coin flip landing on L (probability $\frac{1}{2}$ given $M \neq L$):

$$P(\text{both} \mid \text{distrust}) = \frac{1}{2} \sum_R P(R) P(S_{\text{mi6}} = L \mid R) P(M \neq L \mid R)$$

Case $R = L$

$P(S_{\text{mi6}} = L \mid R = L) = p_f$, $P(M \neq L \mid R = L) = 1 - p_t$.
 Contributes: $r \cdot p_f(1 - p_t)$.

Case $R \neq L$

$P(S_{\text{mi6}} = L \mid R = X) = \frac{1-p_f}{2}$, $P(M \neq L \mid R = X) = 1 - \frac{1-p_t}{2} = \frac{1+p_t}{2}$.
 Contributes: $(1 - r) \cdot \frac{1 - p_f}{2} \cdot \frac{1 + p_t}{2}$.

$$P(\text{both} \mid \text{distrust}) = \frac{r p_f(1 - p_t)}{2} + \frac{(1 - r)(1 - p_f)(1 + p_t)}{8}$$

(Both formulas were checked against direct Monte-Carlo simulation of the exact game and matched to 4 decimal places.)

5 MI6’s Score in Each Regime

Substituting into $\mathbb{E}[\text{MI6}] = 10\mu - 7P(\text{both})$:

$$\begin{aligned} \text{MI6}_{\text{trust}}(p_t, p_f) &= 10 \left[p_f r + (1 - p_f) \frac{1-r}{2} \right] - 7 \left[r p_f p_t + \frac{(1-r)(1-p_f)(1-p_t)}{4} \right] \\ \text{MI6}_{\text{distrust}}(p_t, p_f) &= 10 \left[p_f r + (1 - p_f) \frac{1-r}{2} \right] - 7 \left[\frac{r p_f(1-p_t)}{2} + \frac{(1-r)(1-p_f)(1+p_t)}{8} \right] \end{aligned}$$

6 Optimizing p_f : the Two Threshold Curves

Both formulas above are **affine (linear) in p_f** — so for any fixed p_t , the best p_f is always a corner: $p_f = 0$ or $p_f = 1$. We simply compute the score at each corner and compare.

6.1 Trust regime corners

$$\text{MI6}_{\text{trust}}(p_f=1) = r(10 - 7p_t), \quad \text{MI6}_{\text{trust}}(p_f=0) = \frac{(1-r)(13+7p_t)}{4}$$

Setting $p_f = 1$ better than $p_f = 0$ and simplifying:

$$r(10 - 7p_t) > \frac{(1-r)(13+7p_t)}{4} \iff 53r - 13 - 7p_t(3r+1) > 0 \iff p_t < \frac{53r-13}{7(1+3r)}$$

6.2 Distrust regime corners

$$\text{MI6}_{\text{distrust}}(p_f=1) = r(6.5 + 3.5p_t), \quad \text{MI6}_{\text{distrust}}(p_f=0) = \frac{(1-r)(33-7p_t)}{8}$$

By the same corner comparison:

$$p_t > \frac{33-85r}{7(1+3r)}$$

These are exactly the two curves plotted in Figure 1 below.

7 The Full Picture

Reading the four quadrants

Top-right ($r > \frac{1}{3}$, $p_t > \frac{1}{3}$) — **Trust regime, informative report.** This is the “normal” case: the report is genuinely useful, and Cheka trusts the message. The blue curve cuts through here: *below* it (the large green majority of this quadrant), $p_f = 1$ is optimal; only in a thin sliver near the top (large p_t , moderate r) does $p_f = 0$ edge it out.

Bottom-right ($r > \frac{1}{3}$, $p_t < \frac{1}{3}$) — **Distrust regime, informative report.** The red curve dives below $p_t = 0$ once r exceeds roughly 0.39, so for most of this quadrant $p_f = 1$ is optimal (green). Only in a thin sliver near $p_t = 0$, for r just above $\frac{1}{3}$, does $p_f = 0$ win instead.

Top-left ($r < \frac{1}{3}$, $p_t > \frac{1}{3}$) — **Trust regime, anti-informative report.** Here the blue curve has already dropped below $p_t = 0$ (it only re-enters $[0, 1]$ once $r \gtrsim 0.245$), so it never appears in this quadrant at all: $p_f = 0$ is optimal everywhere here.

Bottom-left ($r < \frac{1}{3}$, $p_t < \frac{1}{3}$) — **Distrust regime, anti-informative report.** The red curve exceeds 1 throughout this region (e.g. it evaluates to ≈ 1.43 at $r = 0.2$), so the condition $p_t > (\text{curve})$ can never be satisfied: $p_f = 0$ is optimal everywhere here too.

The left half of the plane ($r < \frac{1}{3}$) is therefore uniformly a $p_f = 0$ zone, regardless of p_t — this is the key qualitative fact that drives the $r < \frac{1}{3}$ case below.

8 Optimizing p_t : The Two Final Cases

8.1 Case A: $r > \frac{1}{3}$ (report is informative)

We already showed $p_f = 1$ dominates through almost the entire region of interest. Substituting $p_f = 1$:

$$\begin{aligned} \text{MI6}_{\text{trust}}(p_t) &= r(10 - 7p_t), & \text{valid for } p_t \in \left(\frac{1}{3}, 1\right] \\ \text{MI6}_{\text{distrust}}(p_t) &= r(6.5 + 3.5p_t), & \text{valid for } p_t \in \left[0, \frac{1}{3}\right] \end{aligned}$$

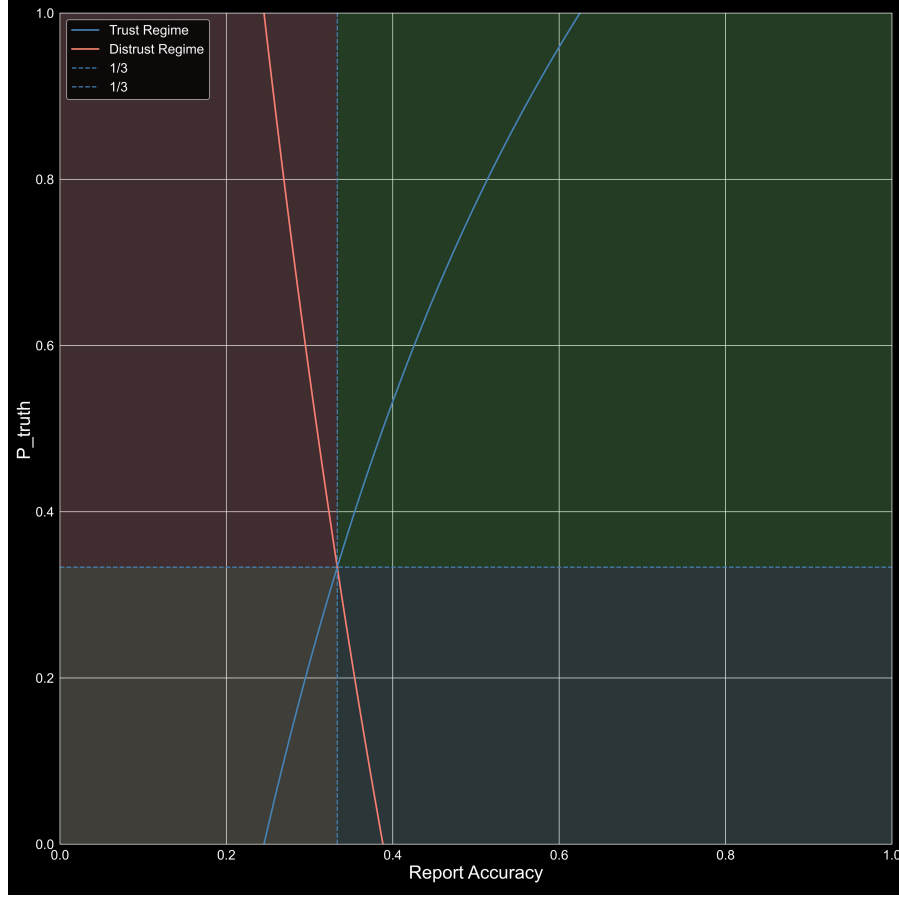


Figure 1: The (r, p_t) plane. The vertical dashed line at $r = \frac{1}{3}$ and the horizontal dashed line at $p_t = \frac{1}{3}$ split the plane into four quadrants. The blue curve is the trust-regime boundary $p_t = \frac{53r-13}{7(1+3r)}$; the red curve is the distrust-regime boundary $p_t = \frac{33-85r}{7(1+3r)}$. Green shading marks where $p_f=1$ is optimal.

The first has slope $-7r < 0$ (decreasing in $p_t \Rightarrow$ best at $p_t \rightarrow \frac{1}{3}^+$). The second has slope $+3.5r > 0$ (increasing $\Rightarrow$ best at $p_t \rightarrow \frac{1}{3}^-$). Both push toward the *same* point from opposite sides:

$$\text{MI6}_{\text{trust}}\left(\frac{1}{3}\right) = r\left(10 - \frac{7}{3}\right) = \frac{23r}{3}, \quad \text{MI6}_{\text{distrust}}\left(\frac{1}{3}\right) = r\left(6.5 + \frac{3.5}{3}\right) = \frac{23r}{3}$$

They meet *exactly* — confirming $p_t = \frac{1}{3}$ is a genuine, continuous peak.

Result — Case A ($r > \frac{1}{3}$)

$$p_t^* = \frac{1}{3}, \quad p_f^* = 1, \quad \text{MI6's optimal score} = \frac{23r}{3}$$

Example ($r = 0.6$): score = $\frac{23(0.6)}{3} = 4.6$. Verified against the exact simulation (with real argmax tie-breaking): 4.606.

Technical note. This result holds robustly once r is not too close to $\frac{1}{3}$ — precisely, for $r > \frac{15}{38} \approx 0.395$. In the narrow window $\frac{1}{3} < r < \frac{15}{38}$, a direct comparison shows the corner $(p_t, p_f) = (1, 0)$ — full honesty, but never follow the report — edges out $(\frac{1}{3}, 1)$ by a small margin. This does not affect either of our representative cases ($r = 0.6$ and $r = 0.2$), but it is worth knowing that MI6’s *own* crossover point is not identical to Cheka’s trust threshold of $\frac{1}{3}$.

8.2 Case B: $r < \frac{1}{3}$ (report is anti-informative)

Here $\frac{1-r}{2} > r$, so the report is *worse* than a random guess. We showed $p_f = 0$ is optimal throughout *both* regimes for $r < \frac{1}{3}$. Substituting $p_f = 0$:

$$\begin{aligned} \text{MI6}_{\text{trust}}(p_t) &= \frac{(1-r)(13+7p_t)}{4}, & \text{slope} + \frac{7(1-r)}{4} > 0 \Rightarrow \text{best at } p_t = 1 \\ \text{MI6}_{\text{distrust}}(p_t) &= \frac{(1-r)(33-7p_t)}{8}, & \text{slope} - \frac{7(1-r)}{8} < 0 \Rightarrow \text{best at } p_t = 0 \end{aligned}$$

Comparing the two corner values:

$$\text{MI6}_{\text{trust}}(p_t=1) = 5(1-r), \quad \text{MI6}_{\text{distrust}}(p_t=0) = \frac{33}{8}(1-r) = 4.125(1-r)$$

Since $5 > 4.125$, full honesty wins.

Result — Case B ($r < \frac{1}{3}$)

$$p_t^* = 1, \quad p_f^* = 0, \quad \text{MI6's optimal score} = 5(1-r)$$

Example ($r = 0.2$): score = $5(0.8) = 4.0$. Verified against the exact simulation: 4.002.

Why the flip? An intuition

When $r < \frac{1}{3}$, the report itself is misleading — each *wrong* city is individually more likely than the report’s suggested city. So MI6 should search *away* from its own report ($p_f = 0$). But MI6 should still relay the report *honestly* ($p_t = 1$): Cheka’s rule doesn’t know the report is unreliable, so full honesty makes Cheka **fully confident** in the very city MI6 has deliberately avoided. MI6 turns Cheka’s own trust into a trap.

9 Summary

Regime	p_t^*	p_f^*	MI6’s Optimal Score
$r > \frac{1}{3}$ (informative report)	$\frac{1}{3}$	1	$\frac{23r}{3}$
$r < \frac{1}{3}$ (anti-informative report)	1	0	$5(1-r)$